

Sandeep Kushawah

Computational Soft Matter · Statistical Physics of the Glass Transition

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RESEARCH INTERESTS	Computational soft matter and the statistical physics of the glass transition. My research uses large-scale molecular dynamics simulations to characterise structure–dynamics relationships in supercooled glass-forming liquids. Current work focuses on developing structural and orientational order parameters, configurational/structural entropy measures, and static and dynamic correlation lengths that connect local packing to slow, heterogeneous dynamics in model binary mixtures near the glass transition.	
EDUCATION	Ph.D. in Physics (in progress) since Feb 2021 Indian Institute of Technology Mandi , India Advisor: Dr. Prasanth P. Jose (Associate Professor) - Thesis area: static and dynamic correlations in Kob–Andersen glass-forming mixtures via molecular dynamics simulation. - Coursework CGPA: 8.0/10 (13 credits).	
	M.Sc. in Physics July 2017 – May 2019 Indian Institute of Technology Delhi , India CGPA: 6.65/10 (80 credits).	
	B.Sc. (Hons.) in Physics July 2014 – June 2017 Shivaji College, University of Delhi , India Minors in Mathematics and Computer Science. Aggregate: 71.33%.	
	PUBLICATIONS Peer-reviewed journal articles [1] Sandeep Kushawah, Devansu Chakraborty, Prasanth P. Jose. Beyond pair entropy: orientational many-body correlations in supercooled glass-forming liquids from a four-point structural entropy. <i>Soft Matter</i>, 2026 (Advance Article). doi:10.1039/D6SM00491A. (first & corresponding author) [2] Sandeep Kushawah, Shubhanjali Pathak, Prasanth P. Jose. Effect of minority composition in a binary glass forming liquid. <i>AIP Conf. Proc.</i>, 2025, 3198(1), 020050. [3] Keshav Thakur, Sandeep Kushawah, Prasanth P. Jose. Relation between local density Debye–Waller factor and density relaxation during glass transition in model systems. <i>AIP Conf. Proc.</i>, 2025, 3198(1), 020051.	
RESEARCH EXPERIENCE	Doctoral Researcher , IIT Mandi Feb 2021 – present Physics of Complex Systems Lab, supervised by Dr. Prasanth P. Jose - Simulate Kob–Andersen 80:20 binary Lennard-Jones glass-forming mixtures using large-scale molecular dynamics. - Develop and analyse static structural/orientational order parameters and configurational entropy measures, and relate them to dynamical heterogeneity and structural relaxation. - Implement and validate custom C/MATLAB analysis pipelines for radial and angular distribution functions and correlation-length extraction. Master’s Thesis – Irreversible Aggregation of Asymmetrical Patchy Particles July 2018 – June 2019 Supervised by Dr. Sujin B. Babu (IIT Delhi) - Simulated asymmetrical patchy colloids using Patchy Brownian Cluster Dynamics to study protein aggregation relevant to neurodegenerative disease. - Visualised configurations using POV-Ray.	

Master's Mini Project – Simulating Plasma using OOPIC

Jan 2019 – June 2019

Supervised by Dr. Vikrant Saxena (IIT Delhi)

- Simulated plasma in a cylindrical chamber using a particle-in-cell model and characterised its behaviour.

**TECHNICAL
SKILLS****Simulation & modelling:** Molecular Dynamics, Molecular Simulation, LAMMPS, GROMACS, VMD, POV-Ray.**Programming & analysis:** C/C++, Python, MATLAB, Bash scripting, $\text{\LaTeX}$.**Computing:** High-Performance Computing (HPC), SLURM, PBS; Linux (CentOS, Ubuntu), Windows; MS Office.**Languages:** Hindi (native), English (professional working proficiency).**SELECTED
PRESENTATIONS
& SCHOOLS**

- Poster, **68th DAE Solid State Physics Symposium**, Bhabha Atomic Research Centre, Mumbai, India.
- Poster, **67th DAE Solid State Physics Symposium**, Bhabha Atomic Research Centre, Mumbai, India.
- **Bangalore School on Statistical Physics XV**, ICTS & Raman Research Institute.
- **Bangalore School on Statistical Physics XIV**, ICTS & Raman Research Institute.

**TEACHING
EXPERIENCE****Teaching Assistant**, IIT Mandi

- Introduction to Molecular Simulation
- Statistical Mechanics
- Statistical Mechanics

Aug 2023 – Dec 2023

Jan 2024 – June 2024

Jan 2025 – May 2025

**PROFESSIONAL
SERVICE****HPC Administration Team**, IIT Mandi

- Managed user accounts and resolved user issues on the institute HPC cluster.
- Assisted users with installation of external scientific software packages.

July 2021 – July 2023

**AWARDS &
HONOURS**

- Best Poster Award, National Science Day 2024, IIT Mandi.
- Qualified CSIR-NET (Lectureship), December 2019 – 96.388 percentile.
- All India Rank 320, JEST Physics 2020 – 95.61 percentile.
- All India Rank 328 / 11,902, IIT JAM 2017 (M.Sc. admission, IIT Delhi).
- All India Rank 547 / 16,990, GATE 2020.

**CERTIFICATIONS
& TRAINING**

- Academic Writing Workshop, IIT Mandi
- Webinar on Science Communication, EURAXESS
- Space Science and Technology Awareness Training (START), ISRO
- MATLAB Onramp, MathWorks (verified)
- Scientific Writing Workshop, ACS Publications

Nov 2024

Oct 2023

Sep 2023

Feb 2022

Sep 2021

REFERENCES**Dr. Prasanth P. Jose** (Ph.D. advisor)

Associate Professor, School of Physical Sciences, IIT Mandi

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